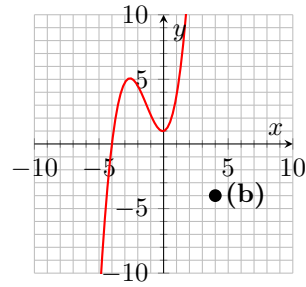


You must show your work to receive credit.

1. Use the graph to complete the following problems.

- [3 pts] Graph and label the point $(-7.5, 4)$.
- [3 pts] Determine the Cartesian coordinates of the point **(b)** on the graph.
- Use the graph to determine the x -int(s) if any of $f(x)$.
- Use the graph to determine the y -int(s) if any of $f(x)$.



Solve the linear equations.

2. $2x + 10 = 4$

3. $3(x - 1) + 13 = -(x + 5)$

4. $9x - (3 + 9x) = 9$

5. $\frac{3x}{2} - x = \frac{x}{10} - \frac{8}{5}$

6. $\frac{x+5}{3} = \frac{5}{3} + \frac{x-8}{2}$

Identify any restrictions on the variable, and solve the rational equation.

7. $\frac{2}{3} = \frac{4}{x} + 1$

Solve the quadratic equations by factoring.

8. $x^2 - 2x = 8.$

9. $4x^2 - 8x - 32 = 0.$

Solve the quadratic equation using the square root property.

10. $(x - 3)^2 - 5 = 7$

Solve the quadratic.

11. $x^2 + 2x = 13$.

Solve the problems

12. [7 pts] When three times a number is subtracted from eight times the same number the result is 20. What is the number?

13. [7 pts] John buys house for \$100,000. If the house increases in value by \$3500 per year. How long until it doubles in value (i.e., how long until it is worth \$200,000)?

14. [Bonus] Jane is playing playing receiver on a flag football team. She runs 10 yards straight down-field, makes a 90°turn to her right, and runs another 5 yards straight towards the sideline. How far is she from her original position.